SENG202 Project Document

Iteration 1

Project: Venture / Barcrawl / Crawler / Ale Trail

Team 5

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Changelog

Keep track of who worked on or reviewed different sections of the document. Include a brief description of the changes made or the responsibilities.

Section	Changes/Responsibilities	Changed by	Reviewed by
Background	Started research into fields of interest related to Alcohol consumption and New Zealand History	Theo	Conrad
Product Overview	Noted down key concepts discussed during meetings and what functional requirements they meet	Theo	Conrad
Current Market	Added basic intro summing key points. To be expanded on	Conrad	
Shareholders	Listed shared holders with ratings and expanded on descriptions	Alistar	Conrad
Current Market	Expanded on main introduction, Created three key weaknesses most competitors face. Strengths and weaknesses of each app type.	Conrad	
Quality Requirements	Created an outline of important requirements to be expanded on earlier	Theo	Conrad
Product Overview	Rewrote key components of the after final deliberation	Theo	Conrad
Stakeholders	Finished the stakeholder table	Alistair	
System context	First iteration of system context diagram and explanation	Neela (27/07)	
Risk Table	Risk assessment table with actions	Alistair	

Business and System Context *(2-3 pages)*

Background ~ Theo

Canterbury is the fastest growing district in Aotearoa, with Christchurch city accounting for 52% of Canterbury's growth. (Stats NZ, 2025) The number of young people moving here is also one of the highest in the country, with a 5.8% annual increase in 15–24-year-olds in 2024. (Burns, 2024) This growth has seen a large increase in restaurants, bars, and nightlife in the city, such as Riverside Market and The Terrace. A popular event, especially among young people, to explore these places is a bar crawl.

A bar crawl, or pub crawl, is a journey to several different bars around a city, sometimes incorporating a theme, competition, or costumes. The bar crawl has likely existed for as

long as pubs have with the earliest organised pub crawl thought to be run by Oxford students in the 19th century. (Pub Crawl Barcelona, 2026) Similar drinking events such as pub golf have been popularised recently by online influencers.

As well as being a fun way to explore a city, going to a variety of bars to have a few drinks is also much safer than other alternatives. House parties, with up to hundreds of young people, especially students, are notorious for becoming chaotic. Police are often called to out-of-control parties at student flats, such as in Riccarton in 2025 where six people were arrested. (Smith, 2025) Similar incidents happen in other young cities, like the infamous incident in 2019 when a student was killed at a flat party in Dunedin. (Sherwood & McNelly, 2019). Venues serving alcohol such as bars, restaurants, and nightclubs must have an on license which has many laws in place to prevent harm from consuming alcohol; minors and intoxicated persons cannot be served, and food and transport options must be available. (Ministry of Justice, 2026). These restrictions make drinking at bars a much safer option compared to visiting house parties.

Especially in a city with so many potential bars, it can be difficult to create a route beforehand that is satisfactory; a pub crawl should be an appropriate distance, not too short or long, have well-spaced stops, and be efficient, minimal backtracking or looping. An application could be made which helps to create a bar crawl route.

Product Overview

The product we will create is an application which helps design a bar crawl within Christchurch City. It makes it easier to create a customised route for a bar crawl, which encourages people to explore the city by foot. It also encourages safer drinking by walking between places that must ensure responsible drinking.

The user should be able to select various parameters to help design the route; minimum and maximum distance, number of locations, start location and end location, bar themes (sports bars, nightclubs), or any particular bars they want to visit. Stops along the way should be suggested, such as water fountains, public toilets, or fast food places.

Users should be able to create an account. This account should be able to be logged back into later. Any routes they create should be able to be saved to their account. Routes should be able to be exported to view on mobile devices. This may be as an image, pdf, link, or similar easily accessible resource.

To ensure legal and responsible drinking, there should be a popup when first opening the app prompting the user to confirm they are 18, the legal drinking age. If they are not, then

they cannot access the application. There should also be links to alcohol help services and safer drinking pages throughout.

Current Market ~ Conrad

What solutions are currently out there that already do something similar? Compare these solutions to your proposed idea. What does your product do better/worse/differently. Does it appeal to the same set of users? You should discuss the pros and cons of each solution and how it compares to your proposal.

NOTES FOR CONRAD TO COMEBACK TO:

How to lie to ATM about how this fits project spec, We are growing the evershrinking population of NZ's youth, Is important for nz to create its own applications for nzers as big bad amerikanas dunno about nz landmarks etc., we hit safety with things like 'drink water here' or 'next bar drink 2 glasses of water before your beer' or 'dont go here with alchol(banned)' or 'Red Card ~ If you get redcarded (kicked out / told you look to drunk to enter) the next place you go to you have to drink water instead of beer' look into PUBBL

Barathon ~ https://play.google.com/store/apps/details?id=com.lunafox.barlog&hl=en_NZ&pli=1

https://www.bar-trek.com

There are currently few applications targeting the niche of 'Route finder bar to bar navigation.' Existing applications such as 'YourPubCrawl' and 'PubCrawl' lack data to perform effectively in the Christchurch area. These focus on aggressive marketing tactics barring essential features from users behind microtransactions and subscriptions. Another app ‘Bar-Trek’, is community based and relies solely on its personal database for reviews, and locations. All apps lack a distinct connection with New Zealand culture and fail to highlight significant features along the walk. They all fail to promote safe drinking habits and safe drinking practices.

[CONRAD COMEBACK ~ Refine why our app is better, what will we do?]

Key Points:

1. Lack Of Support for Christchurch Specific Locations

Current focused applications lack the knowledge of local Christchurch bars focusing their attention on largely populated areas such as USA and Europe. While Google Maps shines at specific locations and distances, it lacks key information such as venue requirements and weekly specials.

2. Safe Alcohol Consumption

Apps primarily focus on the 'consumption' of the alcohol, and not the social aspect. A focus on inebriation and venue-hopping serves as a dichotomy in the Sale and Supply of Alcohol act 2012 248.1 " The licensee or a manager of any licensed premises who sells or supplies alcohol to an intoxicated person commits an offence." Your score is entirely dictated by the number of drinks you consume, and success can only occur after the x'th drink. Without guardrails such as sobriety check's or external activities for those who are drunk, they fail to protect their users.

3. Bar Route Calculations

The algorithm employed by larger apps tend to rely on finding the shortest path between a-b-c and neglect all other factors of pedestrian navigation. The shortest path is not always the safest, routes may recommend dangerous areas such as carparks, poorly lit areas and non-policed areas.

Competitors:

1. Google Maps, Apple maps, Waze (Map focused applications)

General navigation apps often have the most up to date information of bar and most efficient route-finding algorithm.

There is no bar-focused auto routing, leaving navigation between feel a chore than a fun game. The navigation system was designed for driving, and can be unreliable when used for walking as routes can sometimes ignore human limitation.

2. PubCrawl, Bar-Trek, YourPubCrawl, PubGolf (Specific Bar-Hopping Apps)

Built specifically around the intention of a pubcrawl, these apps excel in easily finding a bar route that can be tracked, saved and shared. The live-location and mobile accessibility allows for users to update their friends on location, time and the game side of pubcrawl (challenges, total beers drunk, etc.)

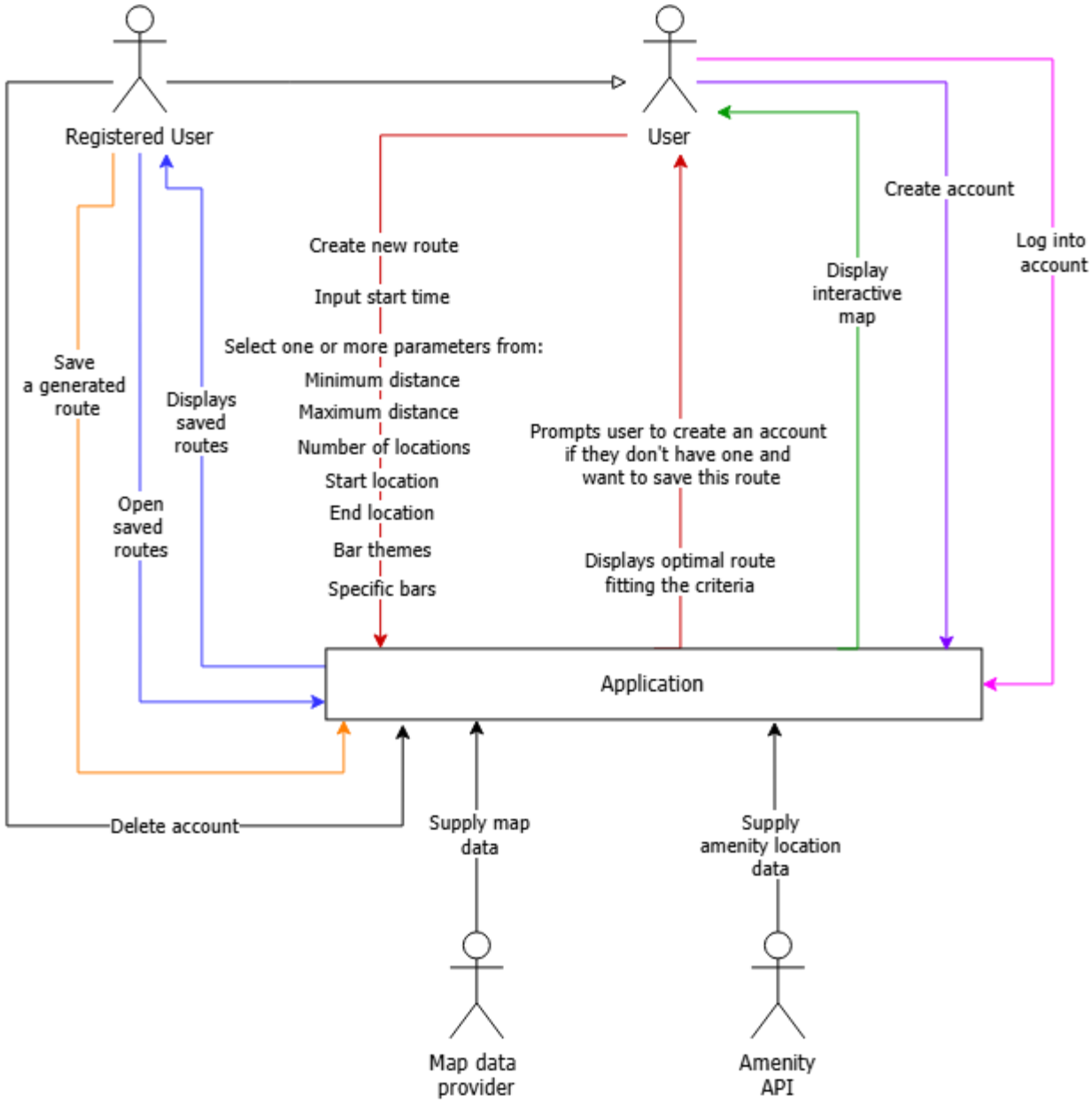
These apps struggle to adapt to change as once a route has started its impossible to mutate. If another joins the group, a bar is closed or unexpected weather events, the app is unable to innovate and change venues to better suit the users. This inability to mutate is not only by application limitations but also data. Larger companies strive for densely populated areas of the world leaving smaller countries data such as Christchurch neglected. These apps also focus on the drinking aspect instead of social. With the

leading intention being alcohol consumption, it leave's little opportunity for those that are sober or severely intoxicated.

System Context ~ Neela

What are the main groups interacting with your final product? Consider the interactions involved, what do they put into the system, what do they get out of it? Demonstrate this with a diagram and discuss the main points. Consider the main features and interactions that will occur between the system and users or other services.

First system context draft



There will be two main groups interacting with the system, users and external data providers.

Users, upon opening the app, will be prompted to either sign in, create an account, or continue as a guest. The application will then load the interactive Christchurch map interface, displaying bar locations. Users will have the option to create a route, in which they will set a starting time and fill out one or more of the following fields: minimum and maximum distance, number of locations, start location and end location, bar themes, particular bars.

The system will use these parameters to calculate an appropriate route fulfilling their specifications. If there are large walking distances within the route, nearby food places, drinking fountains and public bathrooms will be shown on the map, and the user will have the option to add these to their route. The user will then be presented with the route details (walking time, walking distance, locations) and a visual representation of the route overtop the map interface. The user can click a ‘regenerate’ button, which will generate a new route with the same parameters. An option to save the route will then become available. If the user selects this and is signed in, then the system adds a record of the route to the user’s account. If the user is unregistered, they will be prompted to sign in or create an account. Registered users will have the option to view and load their saved routes, as well as sign out and delete their account.

External data providers will provide the system with necessary information, the map data provider giving the map data of Christchurch, and amenity APIs will provide location information for amenities such as drinking fountains and public bathrooms.

Stakeholders *(1 page table, ½-1 page discussion) ~ Alistair*

ID	Name	Description	Priority	Concerns
SH_1	End users	These are the people who are going to use the service provided by the app. Individuals who are 18+ both tourists and locals.	10	They are the primary users of the app so the product's usefulness relies on them. If it doesn't serve them well, the app is unsuccessful.
SH_2	Mana Whenua	The local Māori iwi / hapu (Ngāi Tahu / Ngāi Tūāhuriri) who hold authority and take responsibility over the Christchurch area.	4	Respectful representation of the whenua and wāhi tapu. Creating routes that don't go through significant sites or provide a user warning about the importance. They also care about the wider alcohol related harm in the community.
SH_3	Pub owners / operators	Licenced venues that appear as candidates	8	Being represented accurately (hours, location) is important for maintaining their brand image. Increased foot traffic and spending is beneficial for the venue to increase revenue. They do also care about not being listed in any way which may misrepresent them or their reputation.
SH_4	Third party API Providers	Map tile, routing service providers, plus any other relevant external services the app relies on.	4	That the app implements their services within their terms and gives correct attribution. Needs are more contractual rather than about the purpose of the product.
SH_5	Christchurch City Council	They are the local authority and the source of the some of the open datasets the app relies on. Responsible for public spaces, licencing and city reputation.	7	Their local data is being used appropriately and the city being represented appropriately.
SH_6	Private transport companies	Ride-share services, taxis and e-scooters such as Uber or Flamingo. Their services may be needed for users to get between venues or home safely.	2	A possible referral or integration opportunity. But they have little stake in whether the product succeeds.
SH_7	Developers	The student team designing, producing and maintain the product.	5	Clear requirements technical feasibility, maintainable and testable code. Aiming to create a high quality product.

SH_8	Residents on routes	People living on or near the streets that generated routes pass along	4	Noise, litter, and disorderly behaviour late at night. An accumulation of disturbance on the streets due to common routing.
SH_9	Alcohol regulatory and support boards	District licencing committee and organisations which aim to reduce alcohol related harm, e.g. Health NZ.	6	Responsible promotion of alcohol, minimisation of harm. Not encouraging binge or high-risk drinking, and compliance with the Sale and Supply of Alcohol Act of 2021
SH_10	NZ Police	Local police responsible for public order, law enforcement and alcohol harm and activity in the city.	5	Public safety, managing intoxication related incidents and crowding. Making sure the app doesn't encourage disorderly behaviour. Keeping drinking in licenced places is favourable to private residences.

List out your stakeholders with a table. Each entry should describe who the stakeholder is and why they care about the system. Determine a scale to rank the priority meeting the stakeholder's needs in terms of the product's success. You should discuss how you made the scale and assess the priorities. You should aim for a minimum of 10 stakeholders, think outside of the box!

Stakeholder Write Up:

The priority column

Use Cases *(2-3 diagrams, about 8 tables)*

Make a use case diagram of how your features will fit into the product. If you find that you have use cases that are quite large, feel free to break the diagram up into smaller parts to make it more readable. For iteration 1, we recommend focusing on 2-3 UC diagrams. You should then define a table with 10 to 15 use cases described with an ID, a name, and a priority. For about 7 of them, with the highest priority, you should define the different scenarios as shown in the full table below. These are general guidelines (may vary for different projects). It is also expected to increase slightly with iterations.

Use Case # *Name of the use case*		**Priority** *How important is it to the product?*	**Actors** *Who wants this feature? (e.g. stakeholders)*
1	Create new route	High	End users
2	Regenerate a route with chosen parameters	Low	End users
3	Save a created route	High	End users
4	View saved routes	High	End users, NZ Police
5	Load route from saved routes	High	End users
6	Create an account	High	End users
7	Delete an existing account	Low	End users, developers
8	Log into existing account	High	End users
9	Log out of account	Med	
10	Export route	Med	End users
11	Delete saved route	Low	End users
12	Browse map and locations	High	End users, pub owners/operators
13	Edit parameters of a loaded route	Low	End users
14	View raw location data	Low	Developers, Third party API Providers,
15	Receive alcohol safety pop-up	Med	Alcohol regulatory and support boards, Christchurch City Council

UC_1	*Generate route*
Priority	*High*

Actors		*App users*
Scenario: Blue Sky	**Precondition**	*A user has logged into an account*
	Flow	*What should happen when the use case is run?*
	Postcondition	*What state is the application in after the use case?*
Scenario: Alternate(s)	**Precondition**	*<precondition if different from blue sky>*
	Flow	*Is there another valid flow when the use case is run?*
	Postcondition	*<postcondition if different from blue sky>*
Scenario: Exceptional	**Precondition**	*<precondition if different from blue sky>*
	Flow	*What happens when something goes wrong?*
	Postcondition	*<postcondition if different from blue sky>*

UC_1		Create new route
Priority		*High*
Actors		*App users*
Scenario: Blue Sky	**Precondition**	*The app has launched and successfully loaded all location data and is displaying bars on the map. A tab with all parameter fields is displayed.*
	Flow	*The user fills in every parameter field with valid inputs and clicks the generate button*
	Postcondition	*A route fulfilling the given parameters is generated and displayed on the map*
Scenario: Alternate(s)	**Precondition**	*Same precondition*
	Flow	*The user fills in the minimum number of needed parameters with valid inputs, leaving the other fields empty, and clicks the generate button*
Alternate	**Postcondition**	*Suitable random values are decided for the empty fields, then a route fulfilling the given parameters is generated and displayed on the map*
Scenario: Exceptional	**Precondition**	*Same precondition*
	Flow	*The user doesn't fill out required fields/uses invalid values, and clicks the generate button*
	Postcondition	*Error messages pop up beside empty required fields and fields with invalid inputs*

Quality Requirements (1 page table, ½-1 page discussion)

ID	Title	Description	Steps to Achieve	Stakeholder

QR_1	Route generating	Routes should be generated quickly, and include loading animation	New route searches should have a user perceived response time (UPRT) under four seconds, plus loading animation	End user
QR_2	Route loading	Previously saved routes should load quickly	Saved routes should have a UPRT under two seconds, industry standard (SEO Villas Private Limited, 2024)	End user
QR_3	Navigation	Minimal clicks, understandable to navigate application	Maximum three clicks to navigate to any given page in the application, and/or clear click-path (independent judgement)	End user
QR_4	Aesthetics	Application should look aesthetically pleasing	Consistent with our pre-defined style guide (coordinated colour scheme, typography etc.)	End User
QR_5	Visual Accessibility	Application should be able to be seen easily by users	Colours and contrasts should fit WCAG guidelines (W3C, 2025)	End User
QR_6	Māori Inclusion	Application should include Māori language	Māori names should be included alongside all English names	End User, Mana Whenua
QR_7	Readability	Text should be readable on all screen sizes	Application should have UX scaling, to maximise readability	End User
QR_8	Reading accessibility	Text should be easily readable by all end users	Application should follow dyslexia style guides, or have a toggle, so text is readable (Ako Aotearoa, 2023)	End User
QR_9	Account Security	Accounts should be securely stored within our database	Passwords should be encrypted with BCrypt when put in database to ensure security	End User
QR_10	Data Cache	Calls to API should be minimised	Data from API calls should be stored in database or local files to minimise tokens/usages	Developers, API providers

QR_11	Data Accuracy	Data should be accurate	Data should be updated every 7 days, before each weekend	Bar Owners/ Operators
QR_12	Data Terms	Data should be used according to API terms and conditions	Data should be updated at least every 30 calendar days	API Providers

At least 10% of people are estimated to be dyslexic in New Zealand. (Cole, 2021) Approximately 4.5% of people have a colour vision deficiency. (Greenlane Penrose Optometrists, 2019) Having accessibility features means more people are able to use our application and enjoy a safer bar hopping experience. All users will stay on the application for longer if it looks good, is easy to understand, and is easy to use. (UIUX Design School, 2025) This includes the high speed at which routes should load and generate for the user, and the ease of navigation between pages. The easier and more enjoyable it is to use, the longer they stay on the application, and the more likely they are to use it to create a bar crawl route and be safer while out drinking.

Following data privacy regulations means we can continue using Google's API; this includes reloading data regularly which also ensures accurate information about bars, restaurants, and clubs which will encourage bars to be supportive of our application.

Māori name inclusion is very important in Aotearoa, as it helps us to remember and to connect with culture and heritage. (Manatū Taonga, 2023) As well as being a national language, including Māori names will make people feel connected with the culture and encourage learning about local Māori heritage.

Key Driver Analysis (1 page table, ½-1 page discussion)

A key driver analysis is used to relate our quality requirements back to the stakeholders we identified to better understand which requirements are important and by who. You must discuss the process undertaken to determine stakeholder weightings, as well as reflect on the results

Stakeholder	Weight	QR1	QR_X

End users	10		=
Mana Whenua	4		
Pub owners / operators	8		

Third party API Providers	4		
Christchurch City Council	7		
Private transport companies	2		
Developers	5		
Residents on routes	4		
Alcohol regulatory and support boards	6		
NZ Police	5		

Acceptance Tests *(Around 25 tests – 2-3 pages; led by UCs)*

Consider all functionality provided by use cases. Make sure you cover alternate and exceptional flows as well as both functional and non-functional requirements. The steps for an acceptance test should be concrete and explicit. Anyone should be able to run the test with little to no knowledge of the product. Make sure the acceptance criteria is also explicit and concrete (no partial passes). Make sure each acceptance test only assesses ONE feature.`

Note: you may want to display this table on horizontal page(s), which requires to add Section breaks

ID	Use Case	Scenario	Steps	Acceptance Criteria	Criticality
AT_X	*What use cases are the test related to?*	*What is this testing?*	*What are the steps to execute the test (given/when/then)?*	*How can we concretely know the test has passed?*	*How important is the test to your project?*

GUI Prototypes *(5-7 diagrams with discussions)*

Demonstrate how the application and its features are expected to be used. What will the user flow look like? How will features be grouped? Your designs do not need to be accurate

or high quality. You may propose and evaluate several different designs. Focus on and discuss how you are supporting interaction with your use cases and improving usability.

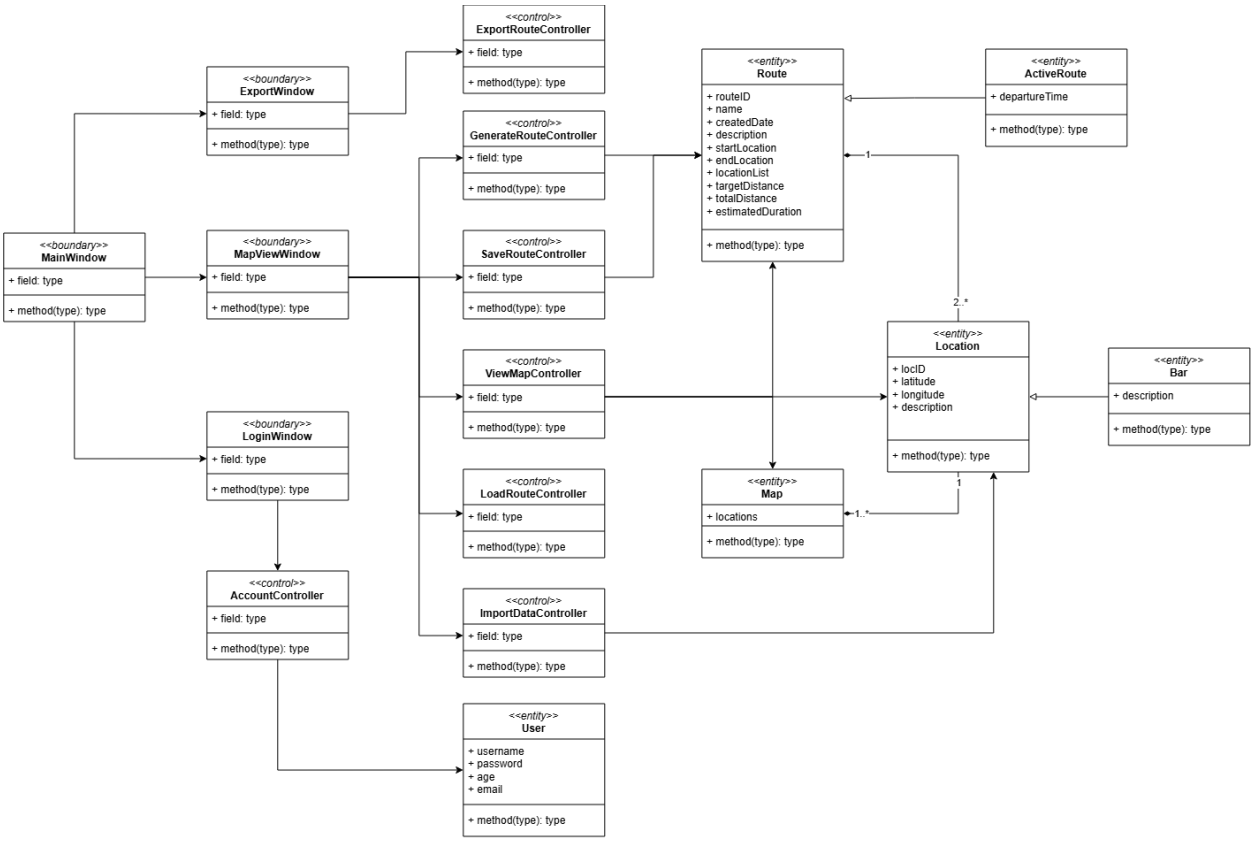
Deployment Model (½ page diagram, ½ page discussion)

Discuss how the application will be deployed with a supporting diagram that shows a simplified overview of deployment considerations and requirements. Focus on specific expected 'shipping' requirements including targeted architecture, external requirements (i.e. Java runtime), connected APIs, shipped files, etc...

Initial UML Class Diagram

(1-2 pages)

Simple class diagram (without attributes and methods) but with relationships among classes. Classes should relate to the main entities, computation steps, and interfaces of the product. You should discuss and justify how your classes are interacting.



Risk Assessment *(1 page table, ½ page discussion)*

Introduce and explain the scales and metrics you are using. Describe how your risk analysis table works. Consider risks of different likelihoods (i.e. things that are very unlikely or very likely, and everything in between, should be considered). Discuss any risks that are still significant after mitigation, is this acceptable? Why or why not

Each risk is measured on two scales ranging from 1-5, 1 being a very unlikely occurrence or negligible impact, and 5 being almost certain to occur or catastrophic consequence. Likelihood measures the probability that the risk will occur during the of project and impact measures the severity of the consequence. Exposure is the product of the two scores and lets us measure where to spend time and effort on mitigating risks. A score of 1-6 (Green) is considered acceptable where we should be conscious of the risk, but no major mitigation strategy is necessary. A score of 8-12 (Yellow) is considered medium risk and to treat where applicable. A score of 13-25 (Red) is considered an unacceptable risk and must be treated before continuing the project.

Table 1- Risk Assessment of Project

ID	Description	Likelihood	Impact	Exposure	Consequences	Prevention/Mitigation	Realized Exposure
RA_1	Venue data is inaccurate or out of date (e.g. opening hours, address, accessibility).	4	3	12	The users are directed to closed or wrong venues. There is then a loss of trust and unfair on pub owners	Timestamp records and show a 'last updated' footnote.	2x3 = 6
RA_2	The app does not comply with the Sale and Supply of Alcohol Act 2012 (e.g. no age acknowledgement or promotes irresponsible drinking)	2	5	10	Legal and ethical breach. Unjustifiable occurrence	Add an age acknowledgement gate. Align project features with aspects of Act.	1x5 = 5
RA_3	Route generation is too slow on CSSE lab computer	3	3	9	Failed performance NFR. Bad impression in demo	Cap number of stops on route. Potentially cache routes.	2x2 = 4
RA_4	Routes or place names ignore Mana Whenua, crosses wahi tapu, or uses incorrect te reo Māori	2	3	6	Cultural offence. Failure of responsibility and stakeholder analysis.	Use correct te reo Māori names. Make route generation conscious of wahi tapu	1x3 = 3
RA_5	Location or personal data is collected or stored, breaching user privacy	2	4	8	Breach of privacy act. There is a loss of user trust and potential data misuse.	Collect minimum data needed. Avoid unnecessary storage. Align with Privacy Act.	1x3 = 3
RA_7	The app design encourages excessive or rapid consumption of	2	4	8	Ethical and reputational harm. Non-compliance with	Have responsible drinking messages / support section.	2x3 = 6

	alcohol, conflicting the NFR				NFR 8. Potential real world harm to users.	Default to walking routes.	
RA_8	A team member becomes unavailable at a crucial time	4	4	16	A part of the project stalls. Knowledge is held by one person. Major time/work pressure on the rest of the team. Especially vital with only having four team members.	Avoid single ownership of code modules. Keep documentation up to date. Pair up for working on high risk / important modules.	3x3 = 9
RA_9	A third part routing API is rate-limited, deprecated, or behind a paywall	3	4	12	Route generation, the main feature, breaks. The map also becomes unavailable	Put routing behind a service interface. Saty inside free token quotas. Potentially cache routes.	2x3 = 6
RA_10	Optimistic estimates and scope cause features to be neglected or left-behind	4	3	12	Incomplete build at the deadline. Requirements unmet. Rushed and low quality code.	Prioritise MVP of core idea. Prioritise tasks. Try to keep a buffer of completed work to allow for time pressure	3x2 =6
RA_11	Poor upkeep of git repository leads to large merge conflicts and a fragmented main branch	4	2	8	Lost project time. Unstable build. Blocked out teammates.	Pull and commit often. CI / CD pipeline to constantly keep on same page. Keep up to date documentation.	2x2 = 4

The table is set up so readers can easily

Project Plan *(1 page table, ½-1 page Gannt chart)*

Break the project down into major and minor milestones. Major milestones should be your course deliverables which you should then break down into a series of smaller minor milestones. For each milestone it should be clear when it is due and what features are expected to be done by that point. The milestones do not have to follow your features exactly but should guide you to delivering the intended product. Remember to include adequate buffer times (more than you think you need), particularly before big deadlines, think about risk management. Have you accounted for a Plan B if things go wrong? What is the Minimum Viable Product at each deadline? You should include the availability of the team (other courses, commitments, work etc.). Consider how this will impact on the amount of work you can complete week to week. Make sure to include a visualization of this plan (e.g. Gantt chart) in addition to your textual descriptions.